

Reference	Title	Authors	Publication Year
[1]	Long Short-Term Memory	Isah Yunusa Umar, K.	2025
[2]	Forecasting Student Drop	Johannes Berens, Ke	2019
[3]	Big Data and Analytics / I	Zohair, L.M.A. et al. (ε	2020
[4]	Addressing student drop	Neema Mduma	2023
[5]	Predicting Student Dropo	Marcell Nagy, Roland	2023
[6]	Prediction of Students' A	Leonard Chukwualuk	2023
[7]	Predicting student dropo	M. Vaarma and H. Li	2024
[8]	Early Prediction of Stude	Carlos Kemper et al.	2020
[9]	Student Dropout Predicti	S. Kim et al.	2022
[10]	Interpretable Machine Le	A. Sharma and P. Kur	2023

Dataset & Size	Methodology / Algorithms	Key Features / Inputs
Student data (attendance, GPA, ac	LSTM Network	Sequence input layer, LSTM layer, fully connected layer
Large dataset (graduates ~1000+,	Regression models, ROC AUC an	Semester-wise enrollment, performance over 4 semeste
Small dataset size modeling / Surv	Educational Data Mining, Predictio	Indirect assessments, major types
Uwezo (Tanzania, 61,340 samples	Machine Learning for highly imbal	Pupil Teacher Ratio, Income, Parent interaction, Latrine
Sensitive University Data	Machine Learning Models	Math score, Chosen subject score, High school GPA, L
Kaggle Dataset (Students' adaptat	Educational Machine Learning Mo	Demographics, Network type, Device, Financial conditio
University student records (Moodle	Predictive Modeling with normalize	Moodle activity trend, Months absent, Accumulated cred
University student records (15k+ r	Random Forest and Decision Tree	Academic performance, Demographics
Online learning platform data	Deep Neural Networks (DNN)	Video watch time, Quiz scores, Assignment completion
Higher education dataset (5k+ rec	XGBoost with SHAP	Socioeconomic status, First-semester grades, Attendan

Performance Metrics	Explainability (XAI)
High accuracy, noticeable clusterin	Not specified / None
Accuracy ranges from 63.47% (1st	None
N/A	None
Dealt with high imbalance (1.6%	None
Evaluated based on feature import	Feature Importance tracking
High accuracy in adaptability predic	Yes (Explainable AI used)
Calculated Spearman correlations	None
Accuracy ~88%	None
Accuracy ~91%	None
ROC-AUC 0.89	Used SHAP for feature contribution

Limitations / Future Work

Did not explore explainability (XAI) for predictions

Static modeling; requires multi-semester tracking for high accuracy

Focuses on small dataset size and general analytics

Highly imbalanced data, needed heavy pre-processing and imputation

Data is highly sensitive and restricted

Focused mainly on online education adaptability rather than absolute dropout

Dependent heavily on active Moodle usage logs

Does not capture temporal changes in student behavior

Lack of transparency in the model's decision making

Focused on static features rather than dynamic multi-semester data